Ch 3.3: Even More Linear Regression

Lecture 7 - CMSE 381

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Wed, Jan 28, 2026

Announcements

Last time:

- 3.2 Multiple Linear Regression

Announcements:

- HW #2 Due Sunday Feb 8.

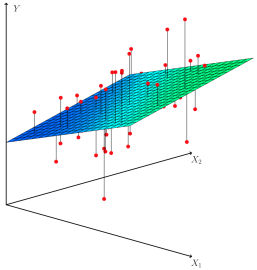
Covered in this lecture

- RSE, R^2
- Confidence intervals and prediction intervals
- Qualitative predictors

Section 1

Continued: Questions to ask of your model

Linear Regression with Multiple Variables



- Predict Y on a multiple variables X

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots \beta_p x_p + \varepsilon$$

- Find good guesses for $\hat{\beta}_0$, $\hat{\beta}_1, \cdots$.
- $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i + \cdots + \hat{\beta}_p x_p$

- $e_i = y_i - \hat{y}_i$ is the ith residual
- RSS $= \sum_i e_i^2$
- RSS is minimized at *least squares coefficient estimates*

Review: Questions to ask of your model

1. Is at least one of the predictors $X_1, \cdots, X_p$ useful in predicting the response?
2. Do all the predictors help to explain Y, or is only a subset of the predictors useful?

- Hypothesis test, F-statistic, at least one β_j is non-zero
- Forward and Backward selection
- RSE, R^2, eyeball test
- Confidence intervals and prediction intervals

Q3

How well does the model fit the data?

Assessing the accuracy of the model

Almost the same as before

Residual standard error (RSE):

$$RSE = \sqrt{\frac{1}{n-p-1} RSS}$$

measure lack of fit;

R squared:

$$R^2 = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

$$TSS = \sum_i (y_i - \overline{y})^2$$

- Proportion in [0,1]; variability in Y that can be explained using X
- 0 means Y not explained by linear regression on X

R^2 on Advertising data

- Just TV: $R^2 = 0.61$
- Just TV and radio: $R^2 = 0.89719$
- All three variables: $R^2 = 0.8972$

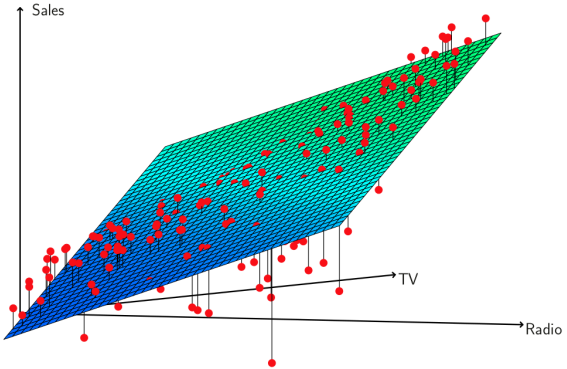
- R^2 always increases with more variables, even if those variables are weakly associated to the response
- Adding another variable decreases RSS on training
- Small increase in R^2 if we include newspaper to other two, so can safely ignore
- Large increase going from just TV to TV and Radio, so likely want to keep that

RSE on Advertising Data

- Just TV: $RSE = 3.26$
- Just TV and radio: $RSE = 1.681$
- All three variables: $RSE = 1.686$

- Similar argument about big jump from TV to TV and radio, but small difference after that
- Even though RSS went down, RSE goes up at the end because increase in p made a difference in the fraction $RSE = \sqrt{\frac{RSS}{n-p-1}}$

If all else fails, look at the data



- Model overestimates sales for cases where either spent only on TV, or only spent on radio
- Underestimates sales where budget was split between the two
- Implies there are some synergy (interaction effects) between the two
- Later (next time?) we'll extend the model to allow for this interaction

Q4

Given a set of predictor values, what response value should we predict, and how accurate is our prediction?

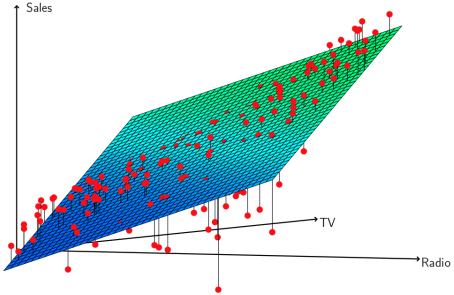
Q4: Making predictions

Given estimates $\hat{\beta}_0, \cdots, \hat{\beta}_p$ for $\beta_0, \cdots, \beta_p$
Least squares plane:

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \cdots + \hat{\beta}_p X_p$$

estimate for the true population regression plane

$$f(X) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$



Confidence vs Prediction Model

Confidence Interval

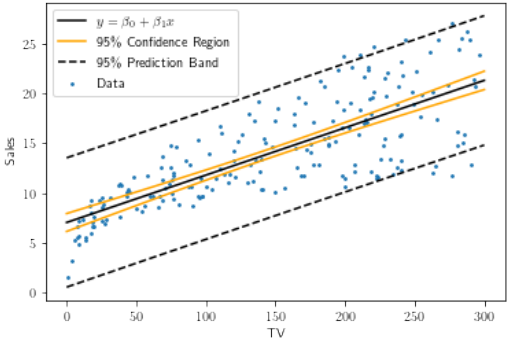
The range likely to contain the population parameter (mean, standard deviation) of interest.

Schrink with more data (n), converge to population regression line/plane.

Prediction Interval

The range that likely contains the value of the dependent variable for a single new observation given specific values of the independent variables.

bound by irreducible error, increase n won't shrink it



Specific to the Advertising Data

Confidence interval: quantify the uncertainty surrounding the average sales over a large number of cities.

Advertising example:
If \$100K is spent on TV, and \$20K on radio, **in each of *n* cities**

95% CI for average sales:
[10,985, 11,528].
Interpretation: if we collect a large number of data sets like the Advertising data set, and we construct a confidence interval for the average sales on the basis of each data set (given \$100,000 in TV and \$20,000 in radio advertising), then 95 % of these confidence intervals will contain the true value of average sales.

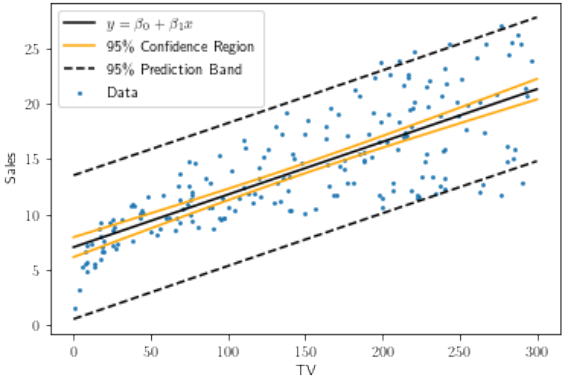
Prediction Interval: quantify the uncertainty in sales for a particular city.

Advertising example:
Given that \$100,000 is spent on TV advertising and \$20,000 is spent on radio advertising in **Gotham City**

95% prediction interval for Gotham:
[7,930, 14,580].
We interpret this to mean that 95% of intervals of this form will contain the true value of Y for this city.

Comparing the two

- Example showing the 1-d regression case
- Fixing TV = \$150, confindence interval for sales is tiny
- High probability the correct model takes a value in there
- OTOH, prediction interval is large
- Need wider band to be confident that a new observation is contained in there



Go take a look at the code under Q4

Review: Questions to ask of your model

1. Is at least one of the predictors $X_1, \cdots, X_p$ useful in predicting the response?
2. Do all the predictors help to explain Y, or is only a subset of the predictors useful?
3. How well does the model fit the data?
4. Given a set of predictor values, what response value should we predict, and how accurate is our prediction?

Test your understanding: PollEv

- Hypothesis test, F-statistic, at least one β_j is non-zero
- Forward and Backward selection
- RSE, R^2, eyeball test
- Confidence intervals and prediction intervals

Section 2

Qualitative Predictors

Reminder: Qualitative vs Quantitative predictors

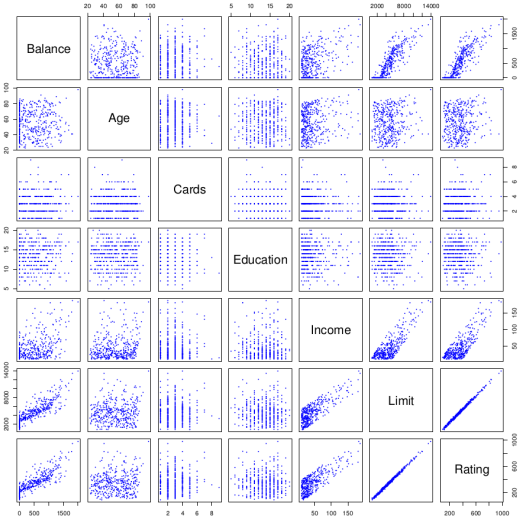
Quantitative:

- Anything with values in either integers or real numbers
- Temperature
- Year (even though integers)
- Weight
- Number of cylinders in a car

Qualitative/Categorical:

- Anything from an unordered set
- Country of origin
- Brand names
- Anything yes/no (married, has a cat)

New data set! Credit card balance



- `own`: house ownership *This is referenced in the text, but isn't in any of the online versions of the data set that I can find*
- `student`: student status
- `status`: marital status
- `region`: East, West, or South *Also referenced in the book, but not in the online data sets*

What if....

... your variables aren't quantitative?

- Home ownership
- Student status
- Major
- Gender
- Ethnicity
- Country of origin

Example

Investigate differences in credit card balance between people who own a house and those who don't, ignoring the other variables.

One-hot encoding

Create a new variable

$$x_i = \begin{cases} 1 & \text{if } i\text{th person is a student} \\ 0 & \text{if } i\text{th person is not a student} \end{cases}$$

Model:

$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_i + \varepsilon_i \\ &= \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{if } i\text{th person is student} \\ \beta_0 + \varepsilon_i & \text{if } i\text{th person isn't} \end{cases} \end{aligned}$$

- β_0 is avg credit card balance among those who are not a student
- $\beta_0 + \beta_1$ is the avg for those are
- β_1 is avg diff in balance

Interpretation

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	480.3694	23.434	20.499	0.000	434.300	526.439
Student[T.Yes]	396.4556	74.104	5.350	0.000	250.771	542.140

- Avg CC balance for non-students \$480.36,
- Students have \$396.46 additional debt, so \$876.83

Model:

$$y = 480.36 + 396.46 \cdot x_{student}$$

Who cares about 0/1?

Old version: 0/1

$$x_i = \begin{cases} 1 & \text{if } i\text{th person is a student} \\ 0 & \text{if } i\text{th person is not a student} \end{cases}$$

Model:

$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_i + \varepsilon_i \\ &= \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{if } i\text{th person is student} \\ \beta_0 + \varepsilon_i & \text{if } i\text{th person isn't} \end{cases} \end{aligned}$$

Alternative version: ± 1

$$x_i = \begin{cases} 1 & \text{if } i\text{th person is a student} \\ -1 & \text{if } i\text{th person is not a student} \end{cases}$$

Model:

$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_i + \varepsilon_i \\ &= \begin{cases} \beta_0 + \beta_1 + \varepsilon_i & \text{if } i\text{th person is student} \\ \beta_0 - \beta_1 + \varepsilon_i & \text{if } i\text{th person isn't} \end{cases} \end{aligned}$$

Note that the predictions will be the same

Qualitiative Predictor with More than Two Levels

`Region`:

	x_{i1}	x_{i2}
South	*1*	*0*
West	*0*	*1*
East	*0*	*0*

Create spare dummy variables:

$$x_{i1} = \begin{cases} 1 & \text{if } i\text{th person from South} \\ 0 & \text{if } i\text{th person not from South} \end{cases}$$

$$x_{i2} = \begin{cases} 1 & \text{if } i\text{th person from West} \\ 0 & \text{if } i\text{th person not from West} \end{cases}$$

$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i \\ &= \begin{cases} \beta_0 + \beta_1 x_{i1} + \varepsilon_i & \text{if } i\text{th person from South} \\ \beta_0 + \beta_2 x_{i2} + \varepsilon_i & \text{if } i\text{th person from West} \\ \beta_0 + \varepsilon_i & \text{if } i\text{th person from East} \end{cases} \end{aligned}$$

More on multiple levels

- Always one less variable than levels
- Level with no dummy variable is called the baseline

	Coefficient	Std. error	t-statistic	p-value
`Intercept`	531.00	46.32	11.464	< 0.0001
`region[South]`	−18.69	65.02	−0.287	0.7740
`region[West]`	−12.50	56.68	−0.221	0.8260

- Balance for the baseline (EAST) is \$531
- South has \$18.69 less than east
- West has \$12.50 less than east
- p-values for dummy variables are large, so can't reject null hypothesis
- No statistical evidence that there's a diff in CC balance in different regions

Do code section on "Playing with multi-level variables"

(If there's time, otherwise next time.)

Next time

CMSE381_S2026_Schedule : Sheet1

Lec #	Date		Topic	Reading	HW	Pop Quizzes	Notes
1	M	1/12	Intro / Python Review	1			
2	W	1/14	What is statistical learning	2.1		Q1	
3	F	1/16	Assessing Model Accuracy	2.2.1, 2.2.2			
	M	1/19	MLK - No Class				
4	W	1/21	Linear Regression	3.1		Q2	
5	F	1/23	More Linear Regression	3.1	HW #1 Due Sun 1/25		
6	M	1/26	Multi-linear Regression	3.2			
7	W	1/28	Probably More Linear Regression	3.3		Q3	
8	F	1/30	Last of the Linear Regression				
9	M	2/2	Intro to classification, Bayes classifier, KNN classifier	2.2.3			
10	W	2/4	Logistic Regression	4.1, 4.2, 4.3.1-3		Q4	